

Joel Sam Mathew

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EDUCATION

Technical University of Munich (TUM)

M.Sc. Informatics – Incoming

Munich, Germany

October 2026

Indian Institute of Technology (IIT) Palakkad

B.Tech. Computer Science and Engineering, CGPA: 9.02/10

Palakkad, India

August 2019 – May 2023

WORK EXPERIENCE

Arista Networks

Software Engineer

Bangalore, India

July 2025 – September 2026

- Built Grafana dashboards on Artifactory access logs to surface upload activity, caller IP, and invoking tool, giving the release team visibility into release-repository usage that hadn't existed before.
- Traced ~100K daily calls to Artifactory (at peak) made under a single shared token back to their originating call sites, then migrated each of them to per-identity authentication via Dex/OIDC token exchange, closing an unaudited access path.
- Implemented CMS-based cryptographic signing for EOS VM image releases, which previously shipped with no way to verify authenticity, and authored the customer-facing [TOI](#) for signature verification.
- Enabled SSH audit logging across release build servers, with events forwarded to CrowdStrike Falcon SIEM for centralized monitoring.
- Engineered a standardized release pipeline for Arista's WiFi product by integrating it with the EOS release framework and automating workflows across the release lifecycle.

ColorTokens Inc.

Member of Technical Staff – II

Bangalore, India

July 2023 – July 2025

- Developed and integrated a dynamic rule engine for policy enforcement within XShield, a zero-trust management platform, reducing infrastructure overhead for key components by over 50% while improving scalability.
- Designed and implemented Layer 3 network policy enforcement within the container micro-segmentation framework using Istio and OPA, enabling fine-grained control over network traffic paths.
- Implemented an opt-in north-south traffic visibility feature, a platform-level switch enabling packet telemetry collection on the server agent, improving control over traffic monitoring and analytics.
- Sole technical point of contact for container micro-segmentation across customer POCs, resolving production issues end to end.

SELECTED PROJECTS

PlexShare

Aug – Dec 2022

- Contributed to the development of a lab session monitoring app in C#, involving features like screensharing, collaborative whiteboard, file uploads, and chat functionalities. Implemented key functionalities in the whiteboard module such as session loading and saving, data serialization and storage, and inter-module networking.
- Stress-tested with a full classroom (~30 concurrent users) screen-sharing and whiteboarding simultaneously in a live trial.

Mail Tag Generator

Jan – May 2023

- Designed and developed a personalized email tagging system combining Latent Dirichlet Allocation (LDA) for topic modeling with GloVe embeddings for candidate tag ranking.
- Implemented the system as a Chrome extension in JavaScript, generating contextually relevant tags for Gmail messages in real time.

Android Malware Detection using GNN

Jan – May 2022

- Developed an Android malware detection approach using Graph Neural Networks trained on API call graphs.
- Applied explainability techniques including GNNExplainer and SubgraphX to identify influential subgraphs and investigate model predictions.

Tiger Compiler

Jan – May 2022

- Developed a compiler for the Tiger programming language in Standard ML using ML-Lex and ML-Yacc, generating MIPS assembly code.

TECHNICAL SKILLS

Languages: Go, Python, C++, C#

Infrastructure & Tools: Kubernetes, Docker, Linux, Ansible, Grafana, Artifactory, CI/CD, Azure Pipelines

AI/ML: TensorFlow, PyTorch

ACHIEVEMENTS

- ColorKudos Award — Individual, for contributions to the Port-Level Zero Trust Enforcement project (ColorTokens, 2024)
- ColorKudos Award — Team, for backend contributions to XShield's policy engine (ColorTokens, 2024)
- Scored 99.67 percentile in JEE Main (1.2M candidates) and 98.11 percentile in JEE Advanced (2019)
- Qualified at the state level for Kerala's Young Innovators Program for an autonomous pesticide-spraying robot (2019)